

NIKISKI HARBOR, AK, USA

WMO: 997382

Lat: 60.683N

Lon: 151.399W

Elev: 26

StdP: 14.68

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.8	(c) 2.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 27.1	(k) 29.7	(l) 24.9	(m) 28.3	(n) 5.8	(o) 90	(p) 0.767

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.9	(c) 64.1	(d) N/A	(e) 62.0	(f) N/A	(g) 60.2	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 7.8	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 30.0	(b) 24.5	(c) 20.7	(e) -6.0	(f) 70.2	(g) 7.9	(h) 4.4	(i) -11.7	(j) 73.4	(k) -16.3	(l) 75.9	(m) -20.8	(n) 78.4	(o) -26.5	(p) 81.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	22.2	23.1	27.1	35.9	45.2	51.4	55.5	55.9	50.2	40.5	28.2	23.8
	DBStd	14.42	11.69	8.81	8.16	5.79	4.48	3.45	2.87	2.92	4.21	5.76	9.48	9.47
	HDD50	4737	861	753	710	424	159	21	0	0	48	295	653	813
	HDD65	9731	1326	1173	1175	874	613	408	293	284	445	759	1103	1278
	CDD50	482	0	0	0	0	11	63	172	182	53	1	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	2	0	0	0	0	0	2	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	9.7	9.6	9.0	9.9	9.7	9.1	10.1	9.9	9.0	9.0	9.7	10.6	10.4
Precipitation	PrecAvg	20.0	1.2	1.2	0.8	0.7	0.7	1.1	1.9	2.6	3.5	2.9	1.7	1.7
	PrecMax	26.9	3.3	3.4	3.9	2.0	2.4	2.2	4.5	5.0	6.7	6.3	4.3	4.4
	PrecMin	13.9	0.3	0.0	0.1	0.0	0.1	0.3	0.6	0.7	1.2	1.2	0.0	0.3
	PrecStd	3.3	0.8	0.8	0.8	0.5	0.6	0.6	1.0	1.2	1.6	1.3	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.9	41.6	44.6	52.2	63.6	66.5	66.1	67.4	60.8	54.4	46.5	42.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	40.7	38.8	41.6	48.6	57.3	61.1	63.1	64.4	58.3	50.8	43.6	39.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	38.2	36.7	39.3	45.8	54.2	58.5	61.3	62.4	56.9	49.3	41.8	38.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	35.9	34.3	37.0	43.3	51.4	56.3	59.8	60.5	55.6	47.9	40.2	36.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	9.3	10.7	10.6	9.5	8.9	7.0	5.9	6.9	7.7	7.6	8.2	8.5
		MCDBR	8.0	8.8	10.2	12.3	14.3	11.9	9.0	10.1	8.7	7.9	6.1	6.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.241	0.288	0.307	0.348	0.371	0.360	0.372	0.358	0.325	0.302	0.253	0.227
	taud		2.149	2.270	2.294	2.273	2.303	2.429	2.428	2.471	2.545	2.458	2.149	1.970
	Ebn at Noon		164	224	260	268	270	276	269	264	256	221	153	105
	Edh at Noon		12	21	28	35	36	33	32	29	22	17	12	9
All-Sky Solar Radiation	RadAvg		93	305	802	1298	1642	1718	1511	1200	779	370	134	47
	RadStd		12	43	93	126	142	124	150	143	112	40	22	6

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		+1.24	+3.76	+3.26	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page